

Dynamic Data Scientist with 6+ years expertise, after my Master degree, in machine learning, deep learning, AI Algorithms development, Data Analysis & Analytics, Quantization, SQL, ETL, AI Agent development, Financial forecasting, Data management, UI and data Visualization, Performance monitoring, Publish/Subscribe using Kafka, NLP, and traditional programming languages including Python and Machine Learning Model (LLMs) with a focus on scalable solutions using Apache Spark Machine Learning (Databricks), Reinforcement Learning, and cross-functional communication/collaboration. I have 1+ years at Amazon working on Data Collection (ETL) and building indices for the extracted data to speed access to them. Built a Python pipeline for Retrieval-Augmented Generation (Generative AI/RAG) using Langchain to generate context-aware marketing content. Additionally, developed a Streamlit web application leveraging OpenAI's GPT-3 model for AI-powered marketing assistance. Developed advanced Generative AI/GAN to manipulate images. Skilled in Azure and AWS data analytics including RDS and DynamicDB, utilizing Redshift, Athena, and QuickSight for scalable pipelines and interactive dashboards. Experienced in all stages of the machine learning lifecycle, from data preparation to deployment, including model inference using BigQuery and Pytorch to testing. Published **bunker-stats-rs** to PyPI, providing **3X-8X faster rolling analytics** and **low-latency anomaly detection** using **Rust** kernels exposed through pyO3 library for Python bindings. Implemented outlier detection (IQR and z-score), scaling methods, KDE, covariance/correlation matrices, and quantile-based binning to improve time-series preprocessing for forecasting and BI workflows

SKILLS

Programming Languages: Python, SQL, R, Perl, JavaScript, Bash

Data Analysis Tools: Microsoft Excel, Tableau, Jupyter Notebook

Cloud Technologies: Amazon QuickSight, Amazon Redshift, Amazon Glue, AWS, Azure, Databricks, Spark

Machine Learning and Deep Learning Frameworks: Pandas, LLM, Scikit-Learn, PyTorch, TensorFlow

Others: Statistics, Git, Docker, Kafka, Microservices, Problem solving, Quantization, Reinforcement Learning, A/B testing

WORK EXPERIENCE

BIG DATA INSIGHTS • Cupertino, CA • 10/2023- Present

DATA SCIENTIST

- Companies usually invest in Business planning; I came up with the idea of predicting future revenues as a tool to help in Business planning. Marketing team validated my assumption and as a result revenue prediction became the core technology for the company. I designed & implemented revenue prediction using Deep Learning
- Researched industry trends and competitor practices in sales forecasting, evaluating diverse methodologies to identify optimal solutions for clients
- Curated historical sales data and applied advanced feature engineering to enhance model performance and accuracy.
- Used statistical models to capture long-term trends and seasonality, while employing k-NN to adjust predictions based on historical analogs and nonlinear patterns.
- Leveraged machine learning expertise with algorithms like ARIMA, SARIMA, and LSTMs to address sales forecasting challenges, developing forecasting models and applications.
- **Leveraged the above experience in designing and implementing the company main product (FutureVision) by implementing N-BEATS neural networks for accurate and robust sales forecasting; significantly improving model performance and prediction accuracy.**
- **Built a Retrieval-Augmented Generation (RAG) pipeline with LangChain and deployed local AI agents for secure, cost-efficient automation of tasks such as SQL query generation, optimization, and context-aware content creation—enhancing both predictive capabilities and data privacy.**
- Implemented a CI/CD **pipeline** with Jenkins for time series projects, automating data preprocessing, model training, and accuracy validation.
- Recently developed a neural network model for a **real-time computer vision system** for apparent age estimation (based on skin image) using transfer learning on EfficientNet and MobileNet architectures with attention mechanisms, GPU acceleration, experiment logging, and benchmarking for speed and MAE accuracy.

SANGU • Sunnyvale, CA • 7/2023- 10/2023 (Part-Time)

DATA ANALYST

- Orchestrated seamless and scalable data importing into a PostgreSQL database, enabling advanced analysis, and significantly contributing to informed decision-making and strategic planning initiatives.
- Leveraged remote access to Linux servers to extract valuable data from log files, employing Perl for efficient data scraping and consolidation, thereby optimizing data retrieval processes.
- Played a pivotal role in migrating crucial customer data into a CRM system, facilitating enhanced relationship management and bolstering customer service endeavors.

AMAZON, LAB 126 • Sunnyvale, CA • 9/2021- 7/2023

DATA COLLECTION SPECIALIST, LAB ENGINEER

- Build scalable data pipelines and ETL workflows using AWS and Google Cloud (BigQuery, Dataflow, Pub/Sub, Redshift, Cloud Storage) as well as forecasting systems and AI-driven applications powered by TensorFlow, PyTorch, JAX, and Apache Spark (Databricks)
- Developed **Python** scripts to elevate data analysis and problem-solving capabilities, encompassing script execution, debugging, and optimization. Conducted meticulous data collection and cleaning from diverse audio devices to ensure exceptional **data quality** and **reliability**.
- Designed and implemented effective data validation strategies and frameworks to identify and address data quality issues, such as **outliers**, **missing values**, and inconsistencies.
- Developed a **BASH script** to systematically validate file naming conventions across multiple folders, ensuring adherence to the prescribed standards and streamlining file organization and management.

BIG DATA INSIGHTS • Cupertino, CA • 8/2019 - 8/21

DATA ANALYST

- Developed a **K-means clustering** algorithm to identify optimal event hosting location based on the population demographics. This helps facilitating data-driven decision-making for successful event planning and execution.
- Applied machine learning techniques to optimize **data migration** processes, resulting in enhancing data availability.
- Researched improving our product competitiveness in a distributed system, with focus on minimizing response-time. Worked closely with the development team to incorporate my findings.
- Conducted competitive analysis to ensure the sustained competitiveness of the product, analyzing market trends, customer needs, and industry benchmarks to inform strategic decision-making and drive product enhancements.

LCI/VERISK • Burlingame, CA • 12/2018 – 8/2019

Operations Analyst

- Developing and implementing **SQL** programs to efficiently manage and modify **database** tables, ensuring optimized **data processing** and seamless integration within the data science workflow.
- Effectively employed proprietary software to perform comprehensive audits of client records, ensuring meticulous examination of data integrity and facilitating actionable insights
- Collaborated with cross-functional teams, leveraging advanced **MS Excel** skills and pivot tables, to analyze and manage data effectively for **reporting** purposes, contributing to the overall success of projects.

EDUCATION

- **Data Science Certificate with Specialization in Machine and Deep Learning**,
Springboard: 6/2020-11/2021
- **M.B.A. in Business Analytics and Technical Marketing**,
St. Mary's College of California: 2015-2018
- **Bachelor of Science in Economics**,
Santa Clara University: 2008-2013. Worked as Information Technology Assistant (1/2011 – 8/2012)

ACADEMIC PROJECTS

Below are some of the projects posted at GitHub (<https://github.com/AdamEzzat1>):

1. Finding Best Location for Sporting events using Clustering ([GitHub Link](#))

- Techniques: K-means clustering
- Tech Stack: Python, Google Colab, pandas, numpy, matplotlib, sklearn, and geopy
- Key Accomplishment: This project used k-means clustering and geography to find the best location for sport events based on demographics.

2. Analyzing Twitter posts for Sentiment analysis and Topic modeling ([GitHub Link](#))

- Techniques: Logistic Regression, Random Forest Classifier
- Tech Stack: Python, Google Colab, Pandas, Numpy, NLTK, Textstat, Matplotlib, Altair, Sklearn, and Keras
- Key Accomplishment: Leveraged Kaggle datasets to scrutinize Twitter posts pertaining to diverse airlines. Employed the tweets to conduct a comprehensive sentiment analysis, coupled with determining the most influential terms contributing to positive and negative sentiments.